

Shengjing Xu

Curriculum Vitae

DRL 3N4A, University of Pennsylvania
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Education

- 2022–present **Ph.D. in Mathematics**, University of Pennsylvania, USA
Expected 2027.
- 2018–2022 **B.S. in Mathematics**, University of Science and Technology of China, China

Honors & Awards

- **Benjamin Franklin Fellowship**, Graduate School of Arts and Sciences, University of Pennsylvania 2022
- **Moruo Guo Scholarship**, University of Science and Technology of China 2021
- **Outstanding Undergrads Golden Awards**, University of Science and Technology of China 2019, 2020

Research Interests

- **Mathematical physics, complex algebraic geometry, and probability.**

Preprints and Publications

- Daebeom Choi and **Shengjing Xu**, Moduli Space of Residue Parameters and Lagrangian Correspondence for λ -Connections, *in preparation*. Draft available upon request.
- Daebeom Choi and **Shengjing Xu**, Colored divisors and Lagrangian correspondence for G Higgs bundle, *in preparation*.
- Jiaoyang Huang, **Shengjing Xu**, and Yulei Xu, Dynamical edge universality and the Airy- β line ensemble, *in preparation*. Draft available upon request.
- Victor Alekseev, Avik Chakravarty, Daebeom Choi, and **Shengjing Xu**, Qin's quasimodularity conjecture for Hilbert schemes of points, 2026, arXiv:2609.33884.
- Xin Sun, Baojun Wu, and **Shengjing Xu**, String equation for conformal loop ensemble on Liouville quantum gravity sphere, 2026, arXiv:2609.17826.
- Jiaoyang Huang and **Shengjing Xu**, Support of Dyson Brownian Motion, 2026, arXiv:2609.01770, *submitted*.
- Avik Chakravarty, Daebeom Choi, and **Shengjing Xu**, Counterexamples to Sato's Weak F -Equivalence Conjecture and a Gorenstein Refinement, 2026, arXiv:2608.18054, *submitted*.
- Baojun Wu and **Shengjing Xu**, Ward identities: a geometric point of view and applications, 2026, arXiv:2608.11550.
- Panagiotis Dimakis, Duong Dinh, and **Shengjing Xu**, Lagrangian correspondences for moduli spaces of Higgs bundles and holomorphic connections, 2026, arXiv:2604.14127.
- Xin Sun, **Shengjing Xu**, and Zijie Zhuang, Annulus crossing formulae for critical planar percolation, 2024, arXiv:2410.04767, *submitted*.

Research Talks

- **Algebraic Geometry Seminar** (*scheduled*)
Stony Brook University, USA, October 14, 2026
- Talk: **Dynamical edge universality and the Airy- β line ensemble**
Special Session on Schramm–Loewner Evolutions and Random Matrix Theory
AMS Fall Eastern Sectional Meeting, George Washington University, USA
October 3, 2026 (*scheduled*)
- Talk: **Holomorphic Lagrangians, Colored Divisors, and Higgs Bundles**
Geometry & Topology Seminar
University of Maryland, USA, September 21, 2026
- Poster Session: **Generalized Baker–Akhiezer divisors and residue parameters**
Summer School: Higher structures, moduli spaces and integrability
University of Hamburg, Germany, August 2026
- Talk: **Holomorphic Lagrangians in Higgs and de Rham Moduli Spaces**
Mathematics-String Theory Seminar
Kavli IPMU, University of Tokyo, Japan, June 25, 2026
- Talk: **Holomorphic Lagrangians in Higgs and de Rham Moduli Spaces**
Caltech/USC Joint Algebra and Geometry Seminar
University of Southern California, USA, April 16, 2026
- Talk: **Holomorphic Lagrangians in Higgs and de Rham Moduli Spaces**
Math-Physics Joint Seminar
University of Pennsylvania, USA, April 2026
- Talk: **Holomorphic Lagrangians in Higgs and de Rham Moduli Spaces**
Differential Geometry Seminar
University of Hamburg, Germany, February 2026
- Talk: **Ward identity: a geometric point of view and applications**
Probability Seminar
Peking University, China, Summer 2025
- Talk: **Liouville quantum gravity sphere**
Probability Seminar
University of Science and Technology of China, China, Summer 2022

Teaching Experiences

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| ◦ MATH 3130 Linear Algebra | Spring 2025 |
| Grader | University of Pennsylvania |
| ◦ MATH 2400 Ordinary Differential Equations | Fall 2024 |
| Teaching Assistant | University of Pennsylvania |
| ◦ MATH 6490 Stochastic Processes | Spring 2024 |
| Grader | University of Pennsylvania |
| ◦ MATH 2400 Linear Algebra | Fall 2023 |
| Grader | University of Pennsylvania |

Academic Service

Referee for:

- Probability and Mathematical Physics
- Advances in Mathematics